

SCOPE · B.TECH PROJECT

Adaptive Physics-Informed Battery Digital Twin for SOH and RUL Prediction

Energy AI · Battery Management Systems · Physics-Informed Neural Networks

TEAM

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SOH · RUL



Project Overview & Use Cases

DOMAIN

Energy AI · Battery Management Systems

TYPE OF DNN

Physics-Informed Neural Network (PINN) + Feedforward DNN

THE CHALLENGE

Li-ion batteries degrade with cycling, temperature and age. Pure deep-learning models need large labeled datasets and can be physically inconsistent; pure physics models are too slow for real-time use.

OUR APPROACH

An Adaptive PINN Battery Digital Twin fuses PINNs, a Transformer Encoder and Incremental Adaptive Learning to continuously estimate SOH and RUL — adapting to new data without full retraining.

WHERE IT APPLIES

Use Case	Impact
Electric Vehicles	Tesla, BYD, Hyundai, Tata EV, MG — predict battery replacement before failure
Battery Swapping Stations	Identify which battery packs should be retired
Solar Energy Storage	Monitor home batteries and predict maintenance
Grid-Scale Storage	Avoid sudden, unplanned battery failures
Drone Batteries	Estimate safe remaining flying time
Space Missions	Battery swaps are impossible — lifetime prediction is critical
Smartphones & Laptops	Sharper degradation estimates than today's health meters

Problem Statement

Lithium-ion batteries power electric vehicles and renewable energy storage, but their performance degrades with charge cycles, temperature, current load and age. Accurately estimating State of Health (SOH) and Remaining Useful Life (RUL) is essential for safety, reliability and efficient maintenance.



Deep Learning Models

- Require large volumes of labeled training data
- May produce physically inconsistent predictions
- Limited generalization to unseen operating conditions

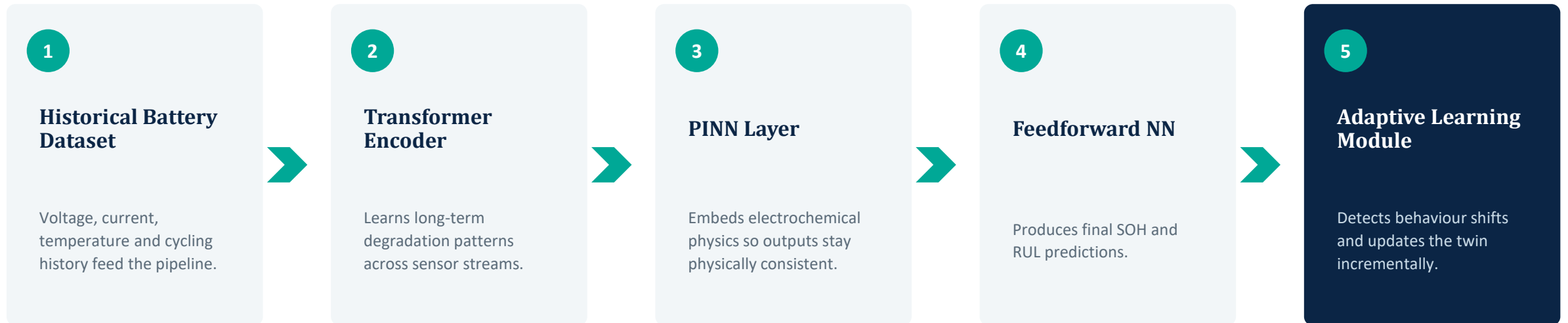


Physics-Based Models

- Computationally expensive to run
- Difficult to deploy for real-time estimation
- Rely on complex electrochemical simulations

Proposed Approach & Architecture

Real-time sensor data, electrochemical battery physics and deep learning combine to estimate SOH and RUL. The Digital Twin continuously adapts to new battery behaviour through incremental learning — no full retraining required.



Base Paper

Physics-Informed Neural Networks for Battery Modeling and State Estimation: A Methodological Review

Renewable and Sustainable Energy Reviews, 2026

WHY THIS PAPER WAS SELECTED

- Comprehensive review of Physics-Informed Neural Networks (PINNs) for battery applications
- Covers SOH, RUL, SOC, Digital Twins, thermal diagnosis and internal state reconstruction
- Explains multiple methods for embedding battery physics into neural networks
- Discusses training stability, domain generalization, uncertainty quantification and interpretability
- Lays out future research directions for intelligent Battery Management Systems

Literature Review & Research Gap

RESEARCH GAP

Existing PINN models are trained offline, adapt poorly to changing battery behaviour, and remain difficult to deploy in real-time BMS. Lightweight, continual-learning methods are still an open problem.

Paper	Key Contribution	Limitation	Our Improvement
Applied Energy (2025)	PINN + Transformer + multi-task learning predicts SOH, RUL and degradation path	Offline training; adapting to new behaviour needs retraining	Adaptive Digital Twin with incremental fine-tuning
Energy (2026)	EM-PINN integrates SEI growth physics for accurate SOH estimation	SOH-only; no Digital Twin or continual adaptation	Combines PINN with a Digital Twin predicting SOH and RUL
Scientific Reports (2025)	Cycle-aware physics-informed framework for RUL prediction	Offline; limited personalization per battery	Continuously updates predictions with new operational data
Complex & Intelligent Systems (2025)	PINN + contrastive learning for low-label SOH estimation	Train-then-deploy workflow; no post-deployment adaptation	Supports incremental learning without full retraining
Proposed Work	Transformer + PINN + Feedforward DNN + Digital Twin + Adaptive Learning	—	Incremental fine-tuning on new battery behaviour, preserving physics constraints

Datasets Used & Rationale

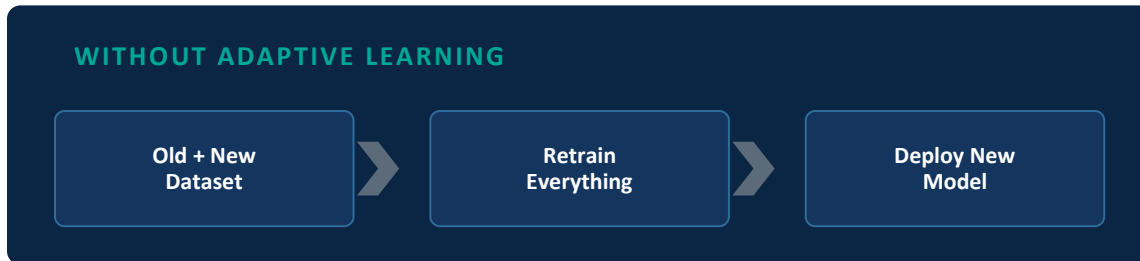
SELECTED DATASETS

Public benchmark battery datasets cited across the reviewed papers, spanning lab-aged, second-life, and vehicle-field conditions, chosen to validate SOH/RUL estimation under diverse chemistries and operating regimes.

Dataset	Scope	Link	Why This Dataset
NASA & Oxford Battery	Lab-aged Li-ion cells; narrow current/temperature range; classic PINN-SOH benchmark	data.nasa.gov (PCoE) / howey-lab Oxford Battery Degradation Dataset	[0] Standard reference set used by our base-paper PINN methodology review to validate degradation-physics constraints
Stanford-MIT-Toyota (Severson)	124 LFP/graphite cells, 72 fast-charging policies, cycled to end-of-life	data.matr.io (Battery Evaluation and Early Prediction project)	[1] Used by the RUL-focused papers for early-cycle degradation trajectories and fast-charging diversity
TJU & XJTU	Multi-chemistry aging data with IC-curve health features across varied cycling protocols	Tianjin University / Xi'an Jiaotong University battery aging repositories	[2] Referenced by the time-series PINN-SOH literature to test cross-dataset transfer of the Adaptive Learning module
CALCE & Panasonic 18650PF	Dynamic charge/discharge cycles, -10°C to 20°C, multi-temperature conditions	calce.umd.edu (Battery Research Group) / NASA Ames Prognostics Center	[3] Provides low-temperature, condition-diverse cycles to stress-test physics-informed robustness
Second-Life LIB (SL-LIB)	84 retired batteries, 21 working conditions, real second-life usage patterns	Self-designed dataset (Bayesian PINN, Applied Energy 2025)	[4] Real-world retired-battery data supports validating adaptive learning on genuinely shifting degradation behaviour

Novel Contribution: Adaptive Incremental Learning

Traditional models are trained once on historical data. When battery behaviour shifts with aging, temperature or charging patterns, the whole model must often be retrained — expensive and impractical for real-time BMS.



HOW IT WORKS

- 1 Monitor real-time sensor data (V, I, temperature, SOC, cycle count)
- 2 Detect changes in degradation behaviour
- 3 Store important new data in a replay buffer
- 4 Incrementally fine-tune the existing model
- 5 Update the Digital Twin — no full retraining

ADVANTAGES

- Eliminates full model retraining
- Adapts to changing battery behaviour
- Personalized prediction per battery
- Physics-consistent predictions via PINN
- Fit for real-time BMS deployment

Applications

1

Electric Vehicles

Predict battery replacement before failure occurs.

2

Solar Storage

Monitor home batteries and predict maintenance needs.

3

Smart Grids

Avoid sudden, costly battery failures at scale.

4

Drones

Estimate safe remaining flying time in-flight.

5

Laptops & Smartphones

More accurate battery degradation estimates.

CONCLUSION

Project Summary

- Proposed an Adaptive Physics-Informed Battery Digital Twin for accurate SOH and RUL estimation.
- Integrated Transformer Encoder, PINN and Feedforward DNN to combine data-driven learning with battery physics.
- Introduced incremental adaptive learning to keep the Digital Twin current without full model retraining.
- Overcomes key limits of pure deep-learning and pure physics-based models with real-time, physically consistent predictions.

Potential impact: Electric Vehicles, Battery Management Systems, Renewable Energy Storage, Smart Grids, Drones and Consumer Electronics.

References

1

Physics-informed neural networks for battery modeling and state estimation: A methodological review

Renewable and Sustainable Energy Reviews, 2026 <https://www.sciencedirect.com/science/article/pii/S1364032126005253>

2

Physics-informed neural network for co-estimation of state of health, remaining useful life, and short-term degradation path in Lithium-ion batteries

Applied Energy, 2025 <https://www.sciencedirect.com/science/article/pii/S0306261925011572>

3

Physical informed neural network for SOH estimation of Lithium-ion battery with electrochemical mechanism

Energy, 2026 <https://www.sciencedirect.com/science/article/pii/S0360544225053964>

4

A cycle-aware and physics-informed framework for battery remaining useful life prediction

Scientific Reports, 2025 https://www.researchgate.net/publication/399255851_A_cycle-aware_and_physics-informed_framework_for_battery_remaining_useful_life_prediction

5

Physics-informed neural network and momentum contrastive learning for battery state of health estimation

Complex & Intelligent Systems, 2025 https://www.researchgate.net/publication/398786756_Physics-informed_neural_network_and_momentum_contrastive_learning_for_battery_state_of_health_estimation

Model Design & Experimental Study

Completed implementation, evaluation evidence, and faculty-ready proof of concept

NASA-first evidence bundle

54 hyperparameter combinations · 3 random seeds · 18 ablation rows · 12 robustness rows

Course rubric focus

Model architecture · Multiple models · Hyperparameter tuning · Experimental analysis · Presentation and AI reflection

Phase II focus: DL model design and hyperparameter tuning

The required work is explicit: design the DL models, tune their hyperparameters, deploy multiple models, and compare results.



Data contract

NASA B0005, B0006, B0007, and B0018 are parsed into ordered discharge-cycle records with voltage, current, temperature, time-gap, and capacity.



Faculty evidence

The run exports final comparison tables, ablations, robustness results, physical diagnostics, figures, and a reflection script.



Model stack

DNN, GRU, Transformer-only, PINN, and adaptive Transformer-plus-physics models run under the same held-out-cell protocol.



What remains a proposal

SEI-PDE physics and broad multi-chemistry validation remain extensions because the NASA POC does not expose every required electrochemical variable.



Online learning

A drift detector triggers low-cost replay-buffer updates as the held-out stream arrives in time order.



Evidence discipline

Reported paper metrics stay separate from new project metrics. The project claims only what the executed protocol measures.

NASA cells and a leakage-safe evaluation protocol

The official NASA PCoE repository provides the first committed dataset for this proof of concept.

Item	Project choice	Why it matters
Cells	B0005 / B0006 / B0007 train · B0018 held out	Battery-level separation tests transfer to a new cell.
Labels	SOH = capacity / nominal capacity × 100 · EOL = 70% SOH	RUL = EOL cycle – current cycle.
Inputs	Voltage · current · temperature · delta_t · cycle index	Measured timing is retained with masks.
Protocol	504 train windows · 132 test windows · 33 initial · 99 stream	Later cycles arrive sequentially for adaptation.
Seeds	7 · 17 · 27	Final tables report mean ± standard deviation.

EOL RULE

70% SOH

HELD-OUT CELL

B0018

CYCLES PARSED

636 total

ORDER

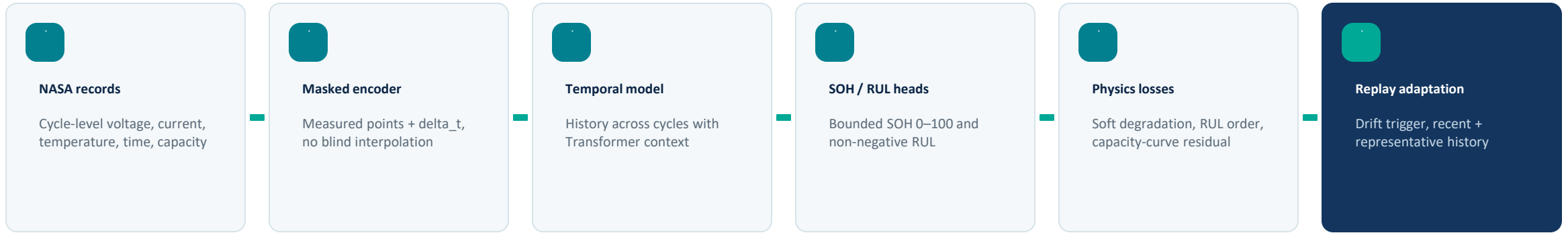
Time-aware

METRIC UNIT

RUL in cycles

From measured cycle to adaptive digital-twin state

The first-stage physics model uses relationships supported by the NASA measurements.



Training objective

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{cons}} L_{\text{cons}}$$

The SEI-growth PDE formulation remains a Stage 2 extension until every variable and parameter is available and verified.

The selected configuration came from the complete grid

Every requested combination ran across seeds 7, 17, and 27 before the final evidence pass.

Grid dimension	Values tested
Hidden size	64 · 128
Learning rate	0.0001 · 0.0005 · 0.001
Physics weight	0.1 · 1 · 10
Replay size	128 · 512 · 2048
Total combinations	54 × 3 seeds = 162 runs

Selected configuration

Hidden size 64

Learning rate 0.001

Physics weight 0.1

Replay size 128

The final evidence runner then used the selected configuration on the full default cycle-window representation.

GRID STATUS

54 / 54 complete

MODEL COUNT

5 comparisons

EVIDENCE MODE

Held-out cell

OUTPUT

JSON + CSV + PNG

The standalone PINN led frozen SOH accuracy

All model rows use the same B0018 held-out evaluation protocol and four training epochs.

Model	SOH MAE	RUL MAE (cycles)	SOH R ²
DNN	11.05 ± 0.25	37.04 ± 6.18	-6.43
GRU	10.52 ± 1.03	36.39 ± 4.95	-1.87
Transformer-only	11.50 ± 0.56	43.98 ± 9.94	-6.57
PINN without adaptation	7.76 ± 1.42	38.15 ± 4.59	-3.86
Adaptive twin	10.05 ± 1.46	38.36 ± 3.39	-4.48

Adaptive twin

10.05 ± 1.46 SOH MAE · 38.36 ± 3.39 RUL MAE

Best frozen SOH row

PINN: 7.76 ± 1.42 SOH MAE

Replay adaptation delivered the low-cost online update

The ablation table shows which components mattered in the ordered B0018 stream.

Variant	Post-shift SOH MAE	Improvement	Work	Events
Proposed adaptive	1.74	80.63%	8.75%	7
No physics	1.23	86.98%	8.75%	7
No Transformer	4.09	50.39%	8.75%	7
No replay	3.97	53.23%	8.75%	7
No drift detection	1.59	81.91%	15.00%	12
Full retraining	1.58	Reference	100.00%	—

TARGET 1

80.63% improvement

TARGET 2

8.75% work

TARGET 3

–10.20% forgetting

DETECTOR

7 updates / seed

The adaptive state remained useful under sensor corruption

Robustness scenarios use the same trained proposal and corrupt the ordered stream before adaptation.

Scenario	Frozen SOH MAE	Adaptive SOH MAE	Updates
Clean ordered stream	8.81	1.73	7
Voltage/current/temp noise	9.02	1.52	7
10% missing samples	8.78	1.51	7
Sensor drift	8.88	1.69	7

Diagnostic	Mean ± SD
SOH bounds violations	0.00%
Negative RUL	0.00%
SOH degradation violation	21.43%
SOH/RUL consistency violation	30.95%
Capacity-curve residual	0.00573

Interpretation

Hard bounds stayed at zero. Soft monotonic diagnostics stayed nonzero, so the result supports a measured proof of concept with calibration limits rather than perfect electrochemical validity.

Proof of experiment mapped to the course rubric

The deck now points to concrete artifacts a reviewer can inspect.

Rubric criterion	What to show	Artifact
Model architecture design	Data → masked encoder → temporal model → physics losses → heads → replay update	HTML report · Section 05
Multiple models	DNN, GRU, Transformer-only, PINN, adaptive twin	final_summary.csv
Hyperparameter tuning	54 combinations across three seeds	tuning/grid.json
Experimental analysis	Ablations, robustness, physical checks, acceptance table	evidence_results.json + PNG figures
Presentation and reflection	6-minute walkthrough and AI pair-programming reflection	video_reflection_script.md

Recording checklist

Show the design diagram · show all five models · open the 54-combination grid · show one ablation and one robustness figure · read PASS status exactly

A tested adaptive POC with a clear next experiment

The strongest evidence is the low-cost online update. The main limitation is the small four-cell dataset.

What the run supports



Adaptation benefit

80.63% lower post-shift SOH MAE than frozen.



Low update cost

8.75% of full-retraining work.

What the run does not prove

- Universal chemistry or temperature generalization
- Perfect monotonic degradation behavior
- SEI-PDE validity without additional variables
- A direct numerical ranking against the four papers

Next experiment

Calibrate physics weights, add longer training, and repeat leave-one-cell-out validation before making a broader scientific claim.

Video demonstration

Empty submission slide for the Moodle, Stream, or Drive video link

VIDEO LINK

PASTE VIDEO URL HERE

Replace this placeholder with the actual uploaded video URL before submitting.

No external video URL was present in the supplied files, so this slide intentionally keeps the link field open.

Equations used in our NASA prototype

These are the target definitions and input contract that are active in the first reproducible implementation.

STATE OF HEALTH

$$\text{SOH}_k = 100 \times C_k / C_{\text{nom}}$$

Measured discharge capacity C_k becomes the supervised health label. Capacity is not fed as a future deployment input.

REMAINING USEFUL LIFE

$$\text{RUL}_k = k_{\text{EOL}} - k$$

RUL is measured in discharge cycles remaining from the current cycle k .

END OF LIFE

$$k_{\text{EOL}} = \min\{k : \text{SOH}_k \leq 70\%\}$$

Primary project threshold. Every RUL result is interpreted against this declared EOL rule.

DEPLOYED INPUT VECTOR

$$\mathbf{x}_k = [V, I, T, \Delta t, k]$$

Voltage, current, temperature, measured time-gap, and cycle index. Irregular timing is preserved with Δt and masks.

Status: implemented and evaluated on NASA B0005/B0006/B0007 → B0018

The active training objective and physical diagnostics

The first-stage POC uses capacity/degradation consistency that the NASA records can support; it does not claim an SEI-PDE residual.

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{cons}} L_{\text{cons}}$$

selected: $\lambda_{\text{phys}} = 0.1$ · $\lambda_{\text{cons}} = 1$ · RUL scale = 250 cycles

DATA LOSS

$$L_{\text{data}} = \text{mean}|\hat{\text{soh}} - \text{soh}| + \text{mean}|\hat{\text{rul}} - \text{rul}|/250$$

Joint SOH and RUL regression with valid-cycle masks.

SOFT PHYSICS LOSS

$$L_{\text{phys}} = \text{mean ReLU}(\Delta\hat{\text{soh}}) + \text{mean ReLU}(\Delta\hat{\text{rul}})$$

Penalizes predicted SOH or RUL increases across ordered cycles; soft because real cells can show local regeneration.

CURVE CONSISTENCY

$$L_{\text{cons}} = \text{mean}|\Delta\hat{\text{soh}} - \Delta\text{soh}| / 100$$

Capacity-curve shape residual between predicted and measured SOH changes.

BOUNDED HEADS

$$\hat{\text{soh}} = 100 \cdot \sigma(z_1), \quad \hat{\text{rul}} = 250 \cdot \sigma(z_2)$$

Output contract: SOH $\in [0, 100]$ and RUL $\in [0, 250]$. The nonnegative-consistency term is an explicit zero-valued future hook.

Diagnostics reported: SOH bounds violation · negative RUL rate · degradation violation rate · SOH/RUL consistency violation · capacity-curve residual

Base paper: general PINN formulation

Transcribed from the methodological review, pp. 4–5. These are representative framework equations, not one new end-to-end model.

(1) NEURAL APPROXIMATION

$$\mathbf{u}\theta(\mathbf{x},t) = \text{NN}\theta(\mathbf{x},t)$$

A network represents a battery state variable such as concentration, voltage, temperature, or SOC.

(2) GOVERNING OPERATOR

$$\mathcal{N}[\mathbf{u}\theta](\mathbf{x},t; \lambda) = 0$$

$\mathcal{N}$ is the physical operator derived from electrochemical, ECM, aging, or thermal equations; λ denotes physical parameters.

(3) RESIDUAL

$$\mathbf{r}\theta(\mathbf{x},t) = \mathcal{N}[\mathbf{u}\theta](\mathbf{x},t; \lambda)$$

Automatic differentiation evaluates the physical-law residual at a point.

(4) PHYSICS LOSS

$$\mathbf{L}_{\text{phys}} = (1/N_p) \sum |\mathbf{r}\theta(\mathbf{x}_i,t_i)|^2$$

Residuals are averaged over N_p physics collocation points.

(5) COMPOSITE PINN LOSS

$$\mathbf{L}_{\text{PINN}} = w_d \mathbf{L}_{\text{data}} + w_p \mathbf{L}_{\text{phys}} + w_b \mathbf{L}_{\text{BC/IC}} + w_r \mathbf{L}_{\text{reg}}$$

Weights balance data fit, governing physics, boundary/initial conditions, and optional regularization.

Project mapping: our $\mathbf{L}_{\text{data}}$ / $\mathbf{L}_{\text{phys}}$ / $\mathbf{L}_{\text{cons}}$ is a smaller NASA-supported specialization of this review framework.

Base paper: electrochemical constraints

These equations are reviewed as physics sources for PINNs; they are not activated in the first NASA POC because the required internal variables are not observed.

(6) SOLID-PHASE DIFFUSION

$$\partial c_s / \partial t = (D_s / r^2) \partial / \partial r (r^2 \partial c_s / \partial r)$$

Fick-type diffusion in a spherical active-material particle; c_s is concentration and D_s is solid-phase diffusivity.

(7) PARTICLE BOUNDARY CONDITIONS

$$\partial c_s / \partial r|_0 = 0, \quad -D_s \partial c_s / \partial r|_{R_s} = J_n$$

Symmetry at the particle center and surface-flux condition at radius R_s .

(8) BUTLER-VOLMER KINETICS

$$j_n = i_0 [\exp(\alpha_a F \eta / RT) - \exp(-\alpha_c F \eta / RT)]$$

Interfacial reaction-current relation; i_0 is exchange current density and η is overpotential.

(9) CHARGE CONSERVATION

$$\nabla \cdot \mathbf{i}_s = -a_s j_n, \quad \nabla \cdot \mathbf{i}_e = a_s j_n$$

Solid/electrolyte current-density balance with specific interfacial area a_s .

Implementation status

Reviewed for Stage 2 extension only · not claimed as an executed SEI/electrochemical experiment

Base paper: ECM state evolution and aging laws

These equations connect macroscopic circuit state, capacity fade, and monotonic degradation constraints.

(10) COULOMB COUNTING

$$dz/dt = -\eta I(t)/Q_n$$

z is SOC, I is applied current, Q_n is nominal capacity, and η is Coulombic efficiency.

(11) RC POLARIZATION

$$dV_p/dt = -V_p/(R_p C_p) + I/C_p$$

First-order equivalent-circuit polarization branch.

(12) TERMINAL VOLTAGE

$$V_t = U_{OC}(z) - IR_o - V_p$$

Open-circuit voltage plus ohmic and polarization terms.

(13) CAPACITY-BASED SOH

$$SOH(k) = Q(k)/Q_o$$

The base review's capacity-health equation is the source for our percentage form $SOH_k = 100Q_k/Q_{nom}$.

(14–15) CAPACITY FADE

$$Q(k) = Q_o - \Delta Q(k) \quad \text{or} \quad Q(k) = Q_o - ak^b$$

General accumulated loss and a semi-empirical cycle-dependent approximation.

(16) MONOTONIC AGING

$$dQ/dk \leq 0 \quad \text{or} \quad Q(k+1) \leq Q(k)$$

The basis for soft degradation constraints; local regeneration can motivate a soft rather than hard penalty.

Project mapping: we use the capacity-health and monotonicity ideas, but not the ECM internal state equations in the first POC.

Base paper: thermal and reaction-rate constraints

The review extends the PINN equation map to coupled thermal fields and temperature-dependent kinetics.

(17) LUMPED THERMAL MODEL

$$C_{th} \, dT/dt = Q_{gen} - hA_s(T - T_{amb})$$

Cell temperature T evolves from heat generation, convection, and ambient temperature.

(18) DISTRIBUTED HEAT CONDUCTION

$$\rho c_p \, \partial T / \partial t = \nabla \cdot (k_{th} \nabla T) + q_{gen}$$

Spatial thermal field equation with density ρ , heat capacity c_p , and conductivity k_{th} .

(19) ARRHENIUS REACTION RATE

$$k_{rxn}(T) = A_r \exp(-E_a / (R_g T))$$

Temperature-dependent reaction-rate constant with activation energy E_a .

Why these are not in the first run

The NASA cycle records expose voltage, current, temperature, time, and capacity, but not every internal concentration, flux, heat-source, or reaction parameter required to claim these residuals.

Scientific boundary

Equations (1)–(19) are the review’s core numbered map; only measurable capacity/degradation constraints are active in our Stage 1 implementation.

Measured model comparison under the held-out-cell protocol

Mean ± standard deviation over seeds 7, 17, and 27; B0005/B0006/B0007 train and B0018 held out.

Model	SOH MAE	SOH RMSE	SOH MAPE	SOH R ²	RUL MAE (cycles)
DNN	11.05 ± 0.25	21.03 ± 0.08	13.74 ± 0.36%	-6.43	37.04 ± 6.18
GRU	10.52 ± 1.03	13.05 ± 0.63	13.69 ± 1.46%	-1.87	36.39 ± 4.95
Transformer-only	11.50 ± 0.56	21.22 ± 0.31	14.42 ± 0.83%	-6.57	43.98 ± 9.94
PINN without adaptation	7.76 ± 1.42	15.81 ± 6.27	9.65 ± 1.53%	-3.86	38.15 ± 4.59
Adaptive twin	10.05 ± 1.46	17.22 ± 5.45	12.82 ± 1.78%	-4.48	38.36 ± 3.39

Readout

Standalone PINN has the lowest mean frozen SOH MAE (7.76 ± 1.42). The adaptive twin’s value is the later-stream improvement and low update cost, not a claim of the best frozen-cell score.

Adaptation targets, ablations, and robustness results

All required experiment families were executed; the acceptance table is the direct proof-of-concept decision record.

Acceptance measure	Mean ± SD	Target	Status
Post-shift improvement	80.63 ± 2.46%	≥ 10%	PASS
Adaptation work ratio	8.75 ± 0.00%	≤ 10%	PASS
Prior-domain forgetting	-10.20 ± 4.47%	≤ 5%	PASS*
Drift events	7 ± 0	> 0	PASS

Scenario	Frozen MAE	Adapted MAE	Events
Clean	8.81	1.73	7
V/I/T noise	9.02	1.52	7
Missing samples	8.78	1.51	7
Sensor drift	8.88	1.69	7

Ablation post-shift SOH MAE

Proposed 1.74 · No physics 1.23 · No Transformer 4.09 · No replay 3.97 · No drift detector 1.59 · Full retraining 1.58

Physical checks: SOH bounds 0.00% · negative RUL 0.00% · capacity-curve residual 0.00573 ± 0.00009 · *negative forgetting means prior-domain MAE improved

What the base review and prior papers actually prove

The project result is new evidence from the executed NASA protocol; literature values remain context because protocols are not directly equivalent.

Source	Experiment / formula focus	Reported result or evidence
0.base_paper.pdf	Methodological review: PINN residuals, electrochemical/ECM/aging/thermal equations, uncertainty, deployment	No single new end-to-end model or one reproducible score; it synthesizes reviewed methods.
Applied Energy paper	Multi-task PINN + simplified Transformer; CV charging features; sensitivity and transfer learning	SOH MAPE 0.75%; degradation-path deviation ≈ 0.01 ; RUL MAE 104 cycles in reported cases.
Energy EM-PINN	SEI-layer-growth PDE with PDE, BC, IC, and SOH losses; XJTU/MIT/HUST; six data-volume scenarios	Reported MAPE as low as 0.873%.
Scientific Reports Bat-T-GNN	Cycle-aware patches, time-aware convolution, Transformer, dynamic GNN, PINN-RUL; ablations and corruption tests	Reported NASA values include RMSE 0.014, MAE 0.007, MAPE 0.008; robustness evidence is central.
Complex & Intelligent Systems	PINN + physics-guided augmentation + momentum contrastive learning on B0005/B0006/B0007/B0018	Compared CNN, BPINN, Informer, XGBoost-ARIMA; added ablation, SHAP, parameter, time, and latency analysis.

Faculty wording

Our defensible novelty claim is replay-buffer adaptation with drift detection and physics-preserving updates; it remains a project claim until replicated beyond this POC.